

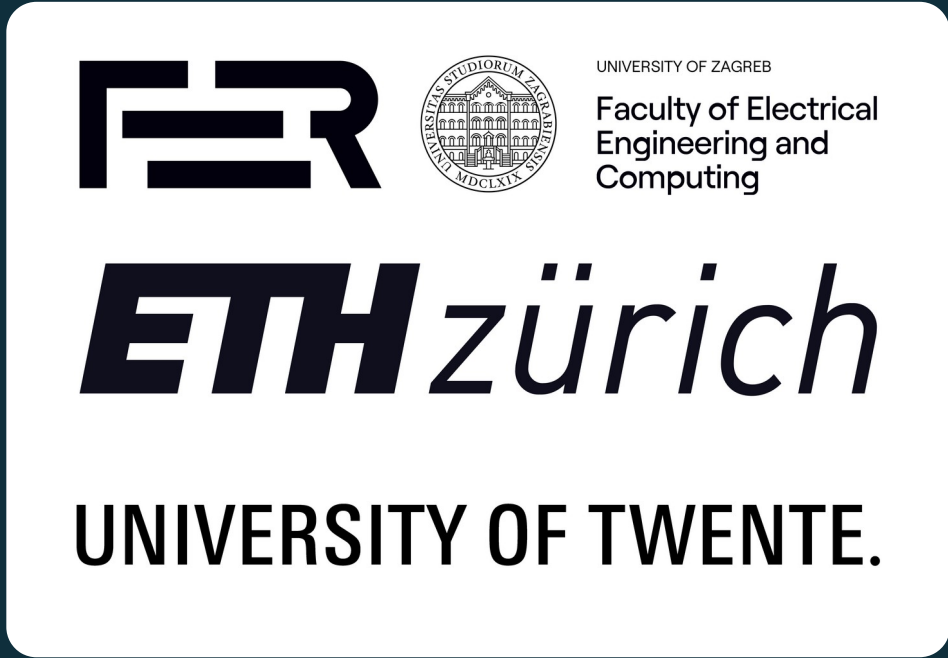
When Adaptation Hurts

Connecting Representational Drift to OOD Failures in MedSAM Fine-Tuning

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Code, configs, checkpoints
github.com/lmSounic/medsam-vpt

Three challenges, studied separately

- **ROI sensitivity.** MedSAM needs a bounding box. Clinical boxes are loose or inconsistent, and Dice follows box quality.
- **Distribution shift.** Accuracy falls when scanner, organ or modality changes.
- **Adaptation cost.** Fine-tuning helps in-domain but is expensive; LoRA and VPT are cheap yet sensitive to data and prompt quality.

This work evaluates adaptation strategy × prompt noise × domain shift jointly, and links the outcome to which internal representations drift (linear CKA vs zero-shot MedSAM).

Setup: adapters and the drift ladder

Method	Params	Updated
Zero-shot MedSAM	0	none (frozen baseline)
Decoder-only FT	4.1M	mask decoder
VPT-shallow	4.1M	10 tokens at input + decoder
VPT-deep	4.2M	10 tokens per block + decoder
LoRA (r = 8, α = 16)	4.4M	encoder + decoder adapters
Encoder-only LoRA	4.1M	encoder adapters, decoder frozen
Full FT	93.7M	image encoder + mask decoder

Train on ISIC 2018 dermoscopy (2,594 images, 80/10/10), 5 epochs, AdamW, Dice + CE, prompt encoder frozen, 3 seeds. Full FT: LR 1e-5, batch 1. PEFT: LR 1e-4, batch 4.

Evaluate along a drift ladder

ISIC 2018 in-domain 1,000 test images	PH² close-OOD dermoscopy 200 images	BUSI far-OOD ultrasound 647 images	CBIS-DDSM far-OOD mammography 362 images
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Prompt protocol. Train with clean boxes, fixed 20 px, or random 0–100 px jitter. Evaluate at 0 / 20 / 50 / 100 / 200 px on 1024² images. 200 px = +280% box area on average, +1,328% on small CBIS-DDSM lesions.

What jitter looks like

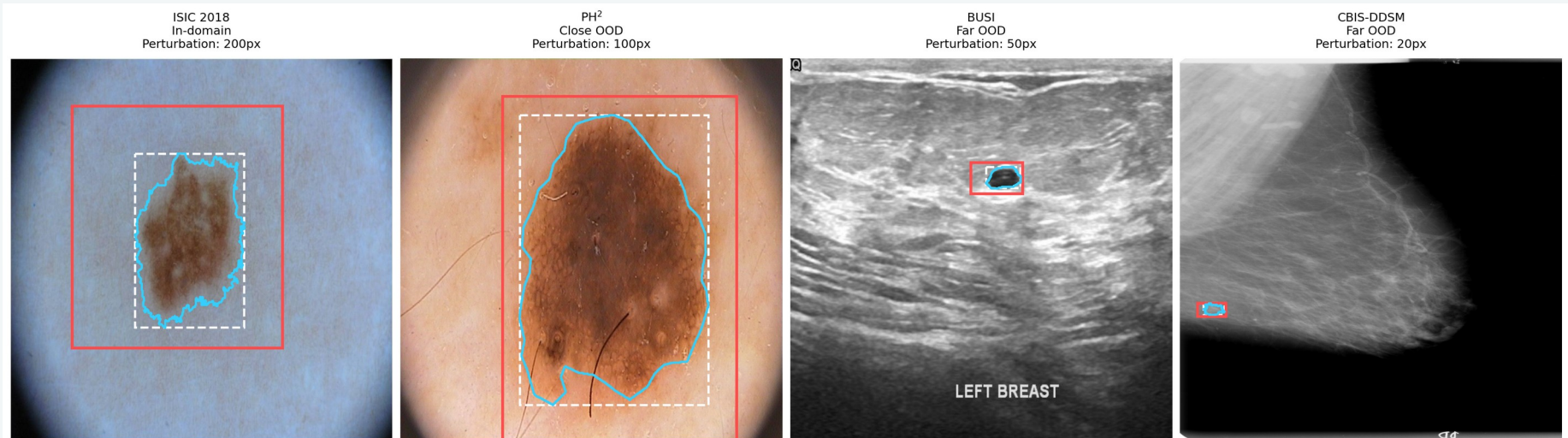


Fig. 1. Tight box (white, dashed), perturbed prompt (red), predicted mask (cyan). Equal pixel jitter is a far larger relative change on small far-OOD lesions.

Result 2 · Clean-box adapters are prompt-brittle

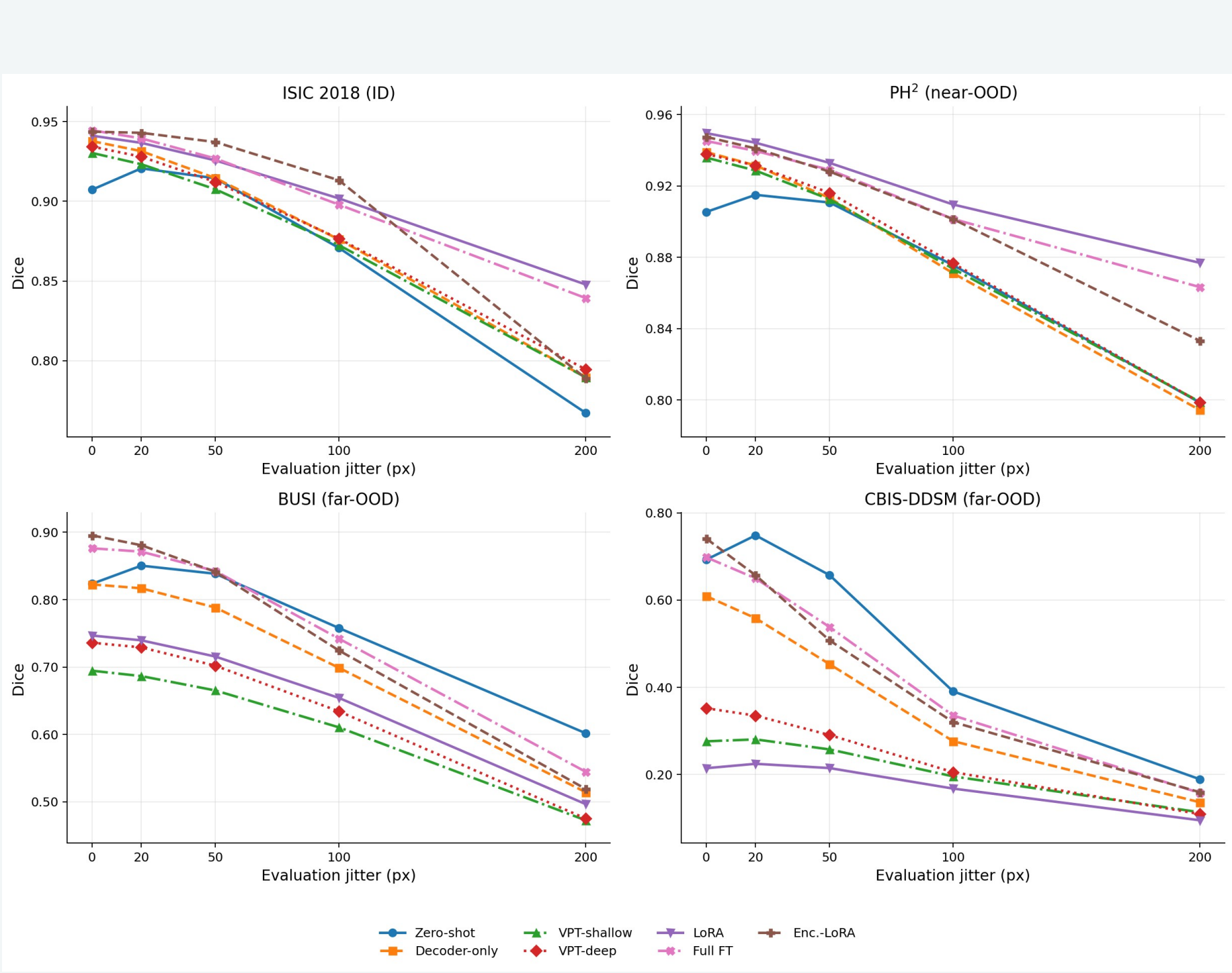


Fig. 2. Dice vs evaluation jitter, averaged over the three training-jitter regimes.

- Zero-shot MedSAM **improves** with 20 px of slack (CBIS-DDSM 0.692 → 0.748) and is the best model on both far-OOD sets once boxes are 100 px off (CBIS 0.391 vs ≤ 0.367 for any adapter).
- MedSAM was pretrained on imperfect boxes; clean-box adapters **unlearn that tolerance**. Even encoder-only LoRA becomes the most jitter-sensitive model in-domain (ISIC, 200 px: 0.645 vs zero-shot 0.767).

Result 1 · Adaptation often hurts far-OOD

Dice with clean boxes, models trained with clean boxes, mean of 3 seeds

Method	ISIC (ID)	PH ² (close)	BUSI (far)	CBIS (far)
Zero-shot	0.907	0.905	0.823	0.692
Decoder-only	0.949	0.947	0.894	0.828
VPT-shallow	0.945	0.943	0.792	0.566
VPT-deep	0.947	0.946	0.802	0.573
LoRA	0.956	0.957	0.780	0.509
Encoder-only LoRA	0.958	0.960	0.908	0.793
Full FT	0.961	0.958	0.901	0.829

- Every adapter beats zero-shot in-domain. LoRA and both VPT variants **fall below zero-shot** on ultrasound and mammography (red).
- **Full FT** gives the best overall trade-off. **Encoder-only LoRA** is the strongest parameter-efficient method: the same adapter as LoRA with the decoder left alone, +0.28 Dice on mammography.

Boundary error: collapse, not overlap loss

HD95 with clean boxes, median [IQR] in pixels at 1024²; lower is better. In-domain and close-OOD medians are 0–7 px for every method.

Method	BUSI (far-OOD)	CBIS-DDSM (far-OOD)
Zero-shot	14 [9, 25]	12 [7, 17]
Decoder-only	4 [2, 17]	4 [3, 6]
VPT-shallow	21 [5, 387]	14 [8, 37]
VPT-deep	25 [6, 394]	18 [10, 359]
LoRA	26 [6, 402]	47 [13, 438]
Encoder-only LoRA	3 [1, 11]	5 [3, 7]
Full FT	4 [1, 13]	4 [3, 6]

- For LoRA and VPT the upper quartile reaches **~400 px**: a quarter of far-OOD cases segment the wrong structure entirely. The Dice gap to Full FT is ~0.12; the boundary gap is more than tenfold.

Result 3 · Train with random 0–100 px jitter

Mean Dice gain over zero-shot: trained with clean boxes → trained with random 0–100 px jitter

Method	In-domain	Far-OOD	Overall
Full FT	-0.006 → +0.034	-0.014 → -0.029	-0.009 → +0.013
Encoder-only LoRA	-0.027 → +0.053	-0.024 → -0.054	-0.023 → +0.007
LoRA	+0.012 → +0.035	-0.142 → -0.228	-0.038 → -0.051
Decoder-only	-0.016 → +0.014	-0.036 → -0.088	-0.023 → -0.022
VPT-shallow	-0.008 → +0.009	-0.127 → -0.230	-0.048 → -0.071
VPT-deep	-0.010 → +0.013	-0.126 → -0.198	-0.049 → -0.058

- Jitter training restores in-domain and close-OOD robustness for every adapter (Full FT at 200 px: 0.712 → 0.839; encoder-only LoRA: 0.645 → 0.789). Fixed 20 px training overfits one noise level.
- It costs far-OOD Dice. The cost is **small** for Full FT and encoder-only LoRA, **large** for LoRA, VPT and decoder-only (LoRA on mammography: 0.509 → 0.214).
- Only Full FT and encoder-only LoRA end up above zero-shot overall.

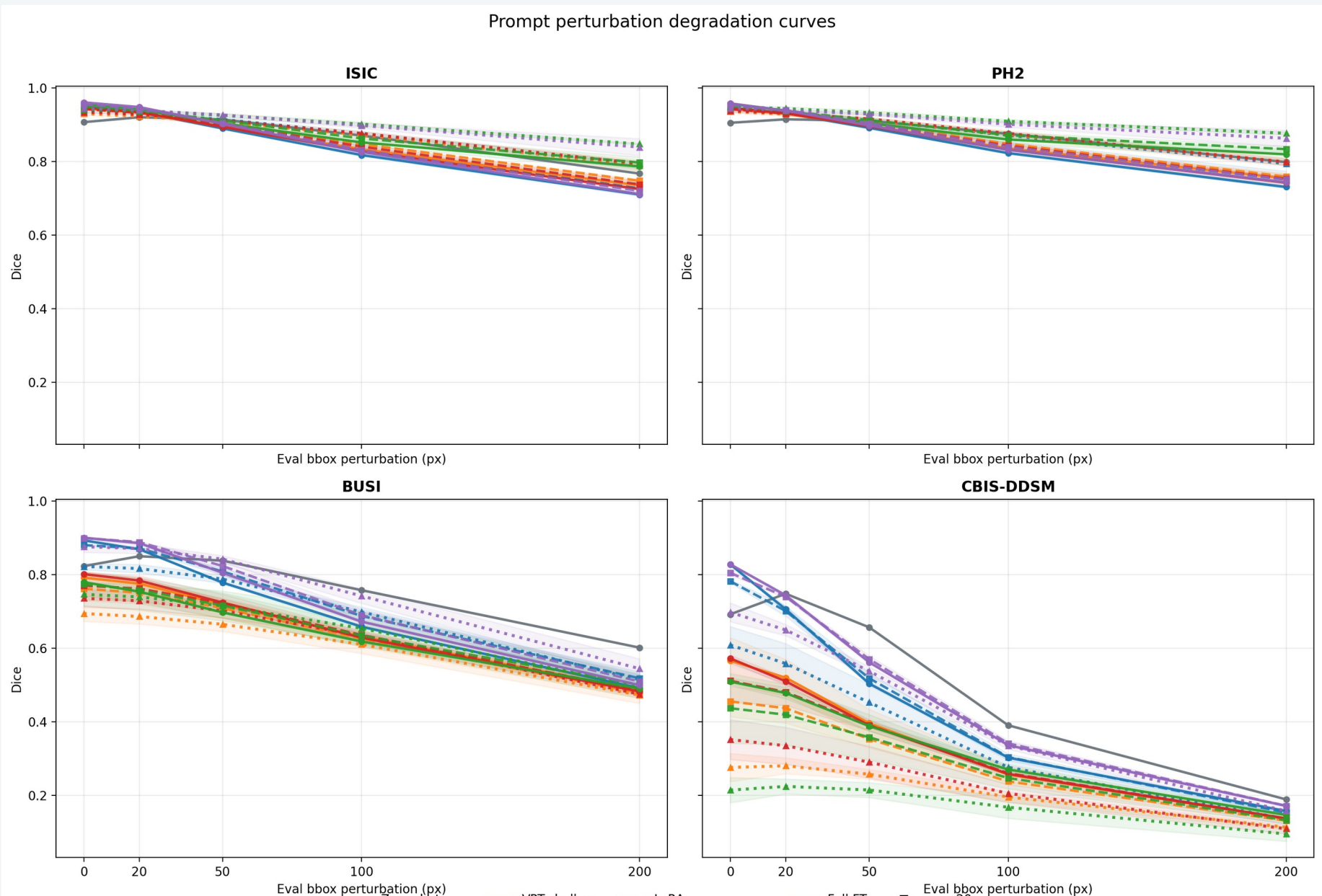


Fig. 3. Dice vs evaluation jitter per training regime (solid: clean, dashed: fixed 20 px, dotted: random 0–100 px). Dotted curves are flattest in-domain and lowest on CBIS-DDSM.

Why does adaptation fail far-OOD?

Linear CKA scores how similar a layer's activations are before and after adaptation (1 = unchanged). We compute it per component against zero-shot MedSAM on the same images and ask whose drift predicts the far-OOD loss, i.e. which part of the model may change.

Spearman ρ between linear CKA (adapted vs zero-shot MedSAM) and Dice gain over zero-shot. n = 3,000 pooled observations; FDR-corrected * q<0.05, *** q<0.001

CKA feature	ID	Close-OOD	Far-OOD
IoU token layer	0.10	-0.02	0.76 ***
Output layer	0.06	0.00	0.76 ***
Decoder layers	-0.08	-0.02	0.73 ***
Upscaled embedding	0.20	0.29 *	0.71 ***
Encoder layers	-0.28 *	-0.28 *	0.09

- **Decoder-side similarity predicts far-OOD gain; encoder similarity does not** (and is weakly negative in-domain).
- Decoder CKA on CBIS-DDSM follows the Dice ranking exactly, from Full FT (0.95) down to LoRA (0.76).
- **Encoder-only LoRA works** because it adapts the encoder to the visual shift while preserving the decoder pathway.

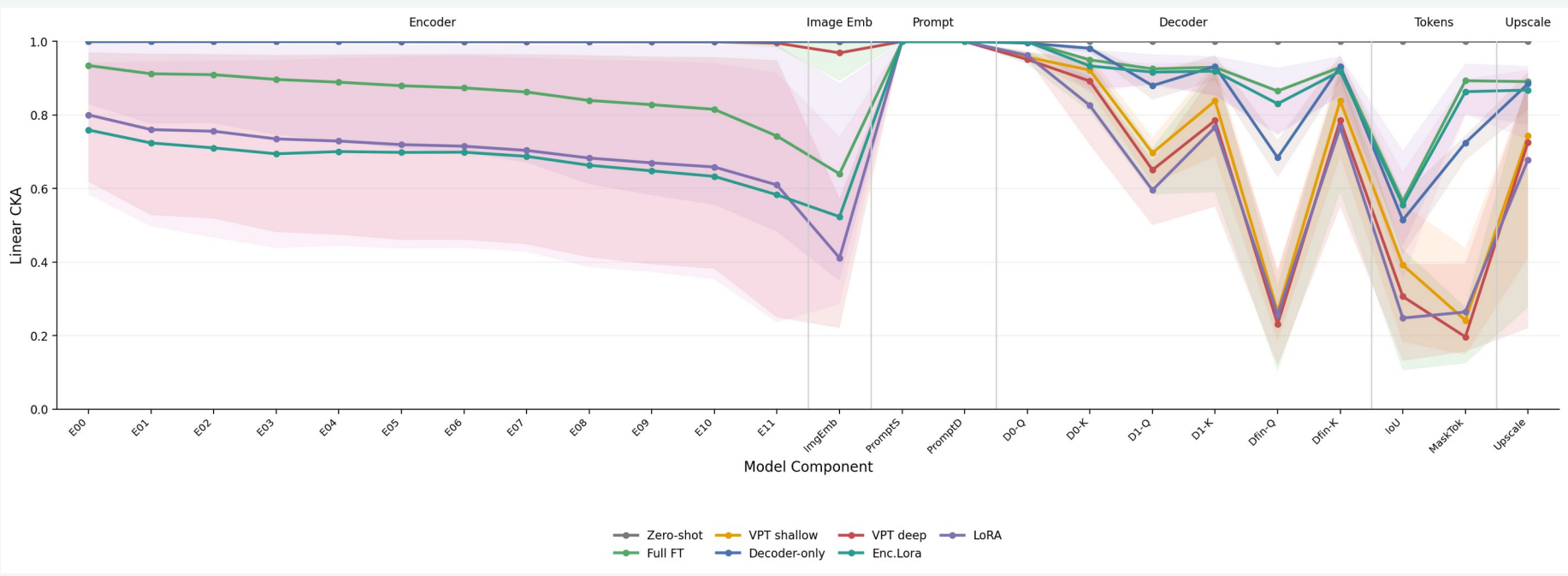
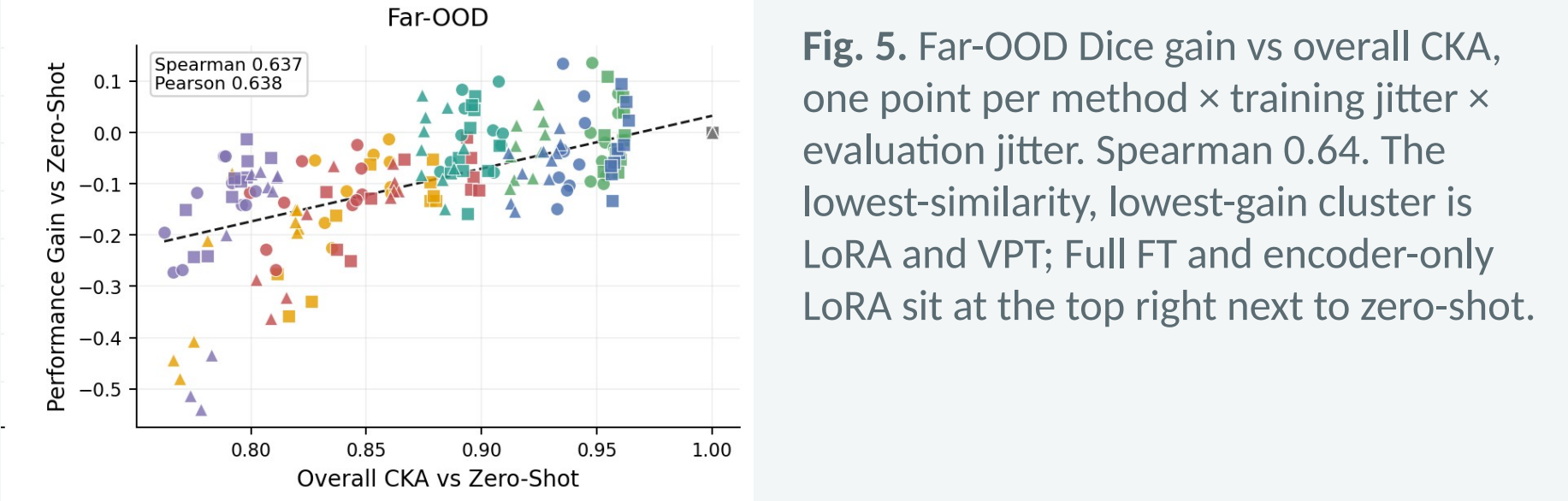


Fig. 4. Linear CKA to zero-shot MedSAM per component (clean boxes). Encoder blocks drift for LoRA and encoder-only LoRA alike; the decoder attention, IoU and mask tokens collapse only for LoRA and VPT.



What we would do with MedSAM now

- With compute available and far-OOD data expected, full fine-tuning at LR 1e-5 gives the best trade-off in every regime.
- On a parameter budget, **encoder-only LoRA** comes closest to Full FT far-OOD with 4.1M trainable parameters.
- For a single target domain with clean prompts, standard LoRA still gives the largest in-domain and close-OOD gains.
- Whatever the adapter, train with random 0–100 px box jitter; fixed jitter overfits one noise level.
- VPT on MedSAM is not worth it. The in-domain gain is negligible and the far-OOD loss is large.
- Under severe prompt noise far-OOD, zero-shot MedSAM is still hard to beat and remains the fallback.

Limitations

- One training domain and two far-OOD targets: modality shift is confounded with lesion-type mismatch. Read the ranking, not the absolute Dice.
- The same pixel jitter is a harder task far-OOD (+152% box area on ISIC vs +1,328% on CBIS-DDSM at 200 px).
- CKA observations are not independent; the effective sample size is well below n = 3,000. Fixed recipe: 5 epochs, LoRA rank 8, 10 prompt tokens.

Take-home

1. **Adaptation can hurt.** LoRA and VPT beat zero-shot in-domain and fall below it on far-OOD modalities, with boundary errors in the hundreds of pixels.
2. **Prompt noise must be trained for.** Random 0–100 px jitter gives the most reliable adapters; only Full FT and encoder-only LoRA end above zero-shot overall.
3. **The decoder is the locus.** Preserving decoder and output representations predicts far-OOD robustness (ρ ≈ 0.73–0.76); encoder similarity does not (ρ = 0.09). Encoder-only LoRA is the CKA-informed adapter.